

BuktiSaham

Task-Based Indonesian Equity Research Orchestration

Combined Product Requirements Document (PRD) and Technical Requirements Document (TRD)

Figure 1. Product object model



Prepared as the approval baseline for actual product and engineering implementation.

0 Document Control and Decision Record

This version supersedes the one-shot research-screen framing. It preserves the Indonesian methodology, local Ollama constraint, free-only data constraint, and deterministic authority boundary while making the task the primary product object.

Field	Decision
Product	BuktiSaham
Version	2.1 task-orchestration blueprint
Implementation checkpoint	Repository v0.2.1
Primary journey	Create Task → Run Saved Method → Observe → Review Evidence → Rerun
Languages	Instant English and Bahasa Indonesia interface switch
AI	Local Ollama, explanation and challenge only
Market data	Free-only yfinance adapter for EOD/delayed MVP
Decision authority	Deterministic Python policy engine
Approval effect	Authorizes implementation against the contracts in this document; it is not an investment recommendation

Key architecture decisions

- A research task is a reusable mandate, not a recommendation.
- Every task update publishes a new immutable configuration version and hash.
- Every run snapshots the selected configuration before it enters the queue.
- A multi-ticker run may publish partial success; successful results are never discarded because one ticker failed.
- AI output is non-authoritative and may never overwrite a deterministic fact, score, scenario, policy result, or hash.
- Localization changes presentation only; it never rewrites stored task evidence or hashes.
- The MVP has no automatic scheduler, authentication, broker connection, or paid-provider fallback.

1 Executive Summary

BuktiSaham is a personal research operating system for Indonesian equities. The user defines a repeatable decision mandate once and runs it whenever fresh evidence is required.

The prior screen executed one ticker immediately and then displayed the result. The underlying backend already had task and run records, but the product did not expose task persistence, configuration identity, run history, progress, or multi-stock orchestration. Version 2.1 aligns the visible product with its intended operating model and adds a persistent English/Bahasa Indonesia interface preference.

Figure 2. End-to-end product journey



Business outcome

- Users can reproduce a method instead of recreating inputs for every check.
- Teams can explain why a recommendation changed by comparing saved runs and evidence, not by guessing which prompt changed.
- A single task can compare a small IDX universe under one consistent horizon and policy.
- Failures become visible operational events and partial results remain usable.

North-star metric

Percentage of active tasks that produce at least two successfully published, comparable evidence packets within 90 days. This measures recurring research utility rather than page visits or one-time analyses.

2 Problem, Users, and Jobs To Be Done

User	Job	Current pain	Success
Self-directed IDX investor	Apply one method repeatedly	Inputs are rebuilt manually and results are difficult to compare	A saved task can be rerun in one action
Equity research analyst	Review several issuers consistently	Per-stock work becomes inconsistent and evidence is scattered	One task runs the same gates across the universe
Product/engineering reviewer	Audit result lineage	UI shows a conclusion without execution trace	Config hash, events, evidence and deterministic hash are visible
Operator	Diagnose failed execution	Background work appears frozen	Stage, percentage, error, and partial state are explicit

Primary job statement

When I want to revisit an Indonesian equity thesis, I want to run the same saved methodology against current available evidence, so I can understand what changed and make a more disciplined research decision.

Problem boundaries

- The product does not promise real-time exchange data.
- The product does not execute orders or optimize a live portfolio.
- The MVP does not claim official IDX data licensing.
- The MVP is single-user local software unless a later security phase adds identity and access controls.

3 Product Principles and Non-Goals

Principle	Implementation meaning
Task before ticker	The home experience starts from mandates and run history, not an isolated symbol form.
Evidence before confidence	Missing or stale data lowers completeness and can hard-block an action.
Configuration is a fact	A run always references an immutable version and hash.
Numbers are deterministic	Python owns calculations, scenarios, and policy actions.
AI is review, not authority	Ollama explains, challenges, and identifies gaps only.
Failures are data	Every queue and worker transition is recorded as a run event.
Free means explicit trade-offs	No billable fallback; availability can be lower and freshness is EOD/delayed.

Non-goals for v0.2.1

- Automated recurring scheduling
- Authentication or shared workspaces
- Portfolio accounting
- Corporate action reconciliation
- Sector-specific valuation
- Automatic run PDF reports
- Regulated personalized advice
- Brokerage execution

4 User Journey and Information Architecture

Figure 3. Workspace information architecture



Area	User question answered	Primary actions
Overview	What is active and what changed recently?	Create task, open task, inspect recent run
Research Tasks	What reusable mandates exist?	Create, inspect, run, later edit/archive
Task Detail	What exact method will execute?	Review thesis, universe, modules, config hash, history
Run History	What happened in every execution?	Inspect status, stage, progress, error, packet
Run Detail	What did the task conclude and why?	Select ticker, inspect scenarios, evidence, Ollama review
Methodology	How are actions produced?	Review deterministic and AI authority boundaries
Settings	How should the workspace be displayed?	Switch English/Bahasa Indonesia without reload

Happy path

- User creates a named task with thesis and one to ten IDX symbols.
- System normalizes symbols, validates the task, publishes config v1, and displays its hash.
- User selects Run saved methodology.
- API rejects duplicate active execution, snapshots the config, stores a queued event, and enqueues work.
- UI polls and shows aggregate stage progress.
- Worker publishes complete or partial packet with per-symbol drill-down.
- User revisits the task later and runs the same or a newly versioned configuration.

5 Functional Requirements - Research Tasks

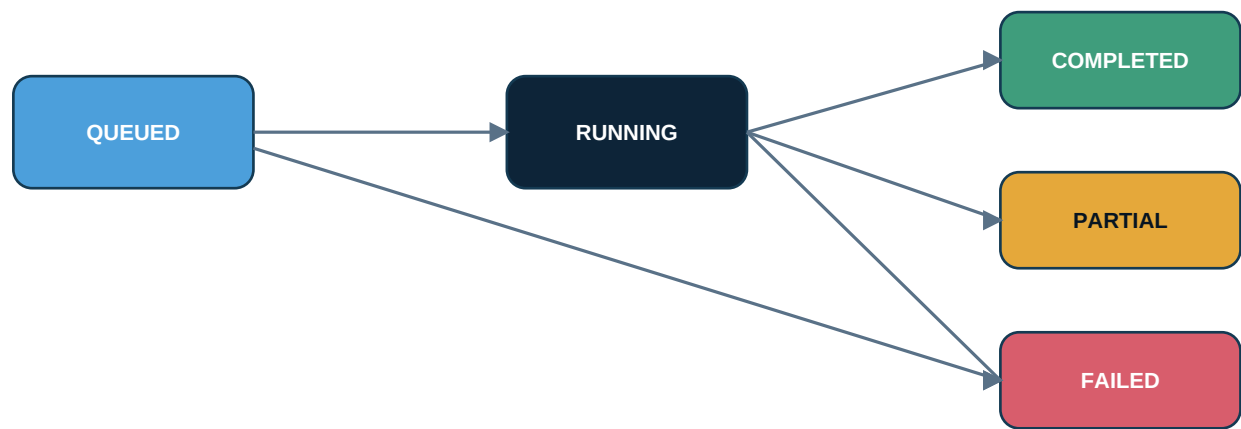
ID	Requirement	Acceptance
TASK-01	Create a task with name, thesis, symbols, horizon, capital context, cadence metadata, and modules	Valid request returns 201, normalized .JK symbols, config version 1 and hash
TASK-02	Allow one to ten unique symbols	Duplicates are removed; zero or more than ten fails validation
TASK-03	Preserve legacy single ticker input	A request with ticker only creates a one-symbol task
TASK-04	List and retrieve active tasks	Responses include latest run summary and config identity
TASK-05	Update task as new config version	Core task ID remains stable while version increments and hash changes
TASK-06	Archive a task	Archive is blocked while a run is active; archived task is omitted from lists
TASK-07	Show explicit MVP cadence boundary	UI states that runs are manual; cadence is metadata only

Task contract

name, thesis, tickers[], horizon_days, capital_idr, cadence, analysis_modules[], status, market_data_mode, provider

6 Functional Requirements - Runs and Orchestration

Figure 4. Run state machine

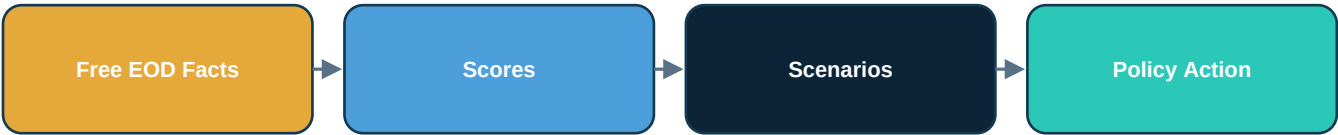


ID	Requirement	Acceptance
RUN-01	Snapshot active config before enqueue	Run contains config version, hash, and full config even if task later changes
RUN-02	Prevent concurrent duplicate task runs	Second start returns HTTP 409 with active run ID
RUN-03	Emit append-only stage events	Polling returns ordered stage, progress, message, and timestamp
RUN-04	Aggregate progress across ticker universe	Progress is non-decreasing and reaches 100 in a terminal state
RUN-05	Continue after one symbol error	Remaining tickers execute; mixed result becomes PARTIAL
RUN-06	Publish immutable task packet	Bundle includes summary, analyses, errors, snapshot and SHA-256 hash
RUN-07	Expose task-specific and global history	User can list all runs or filter by task ID

7 Indonesian Equity Research Methodology

The current methodology is deliberately transparent and provider-constrained. It is suitable as an MVP research screen, not a substitute for licensed data or professional judgment.

Figure 5. Per-ticker methodology



Layer	Current calculation	Decision role
Market history	One year daily close and volume	Technical trend, volatility, drawdown, liquidity
Fundamentals	Available valuation, growth, profitability and leverage fields	Fundamental score and evidence coverage
Technical score	Return, moving-average, RSI and risk signals	Policy input
Liquidity score	Observed volume and tradability proxy	Hard/soft policy gate
Evidence completeness	History depth plus available fundamental fields	Confidence and hard gate
Scenarios	Bear/Base/Bull target prices and probabilities	Expected return and downside
Policy	Configured thresholds and hard blocks	Final research action

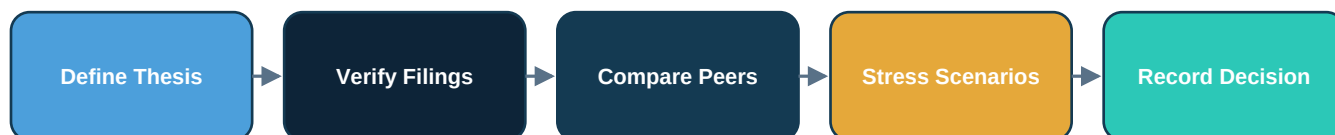
Research actions

- BUY_RESEARCH: evidence and expected return pass configured upside and risk gates.
- HOLD_RESEARCH: evidence is usable but upside is inside hold range.
- SELL_RESEARCH: downside/valuation policy supports reduction research.
- WATCH: confidence, completeness, liquidity, or consistency is insufficient for a stronger action.

7A Pro Tips for Disciplined Stock Research

The goal is a higher-quality, risk-aware decision process - not a promise of the highest return. Use BuktiSaham as a repeatable research assistant, then verify material facts against official disclosures before acting.

Figure 5A. Disciplined research loop



Pro tip	How to apply it	Common mistake avoided
Start with a falsifiable thesis	Write what must be true, the expected horizon, and the evidence that would invalidate the idea.	Changing the story after the price moves.
Check official disclosures	Read the latest audited annual report, interim statements, material information, corporate actions, and issuer presentation. Record each as-of date.	Relying only on price feeds, summaries, or social media.
Use sector-fit metrics	Compare banks with banking ratios, commodities with cycle and cost drivers, and consumer names with margin, volume, and working-capital measures.	Using one valuation multiple across unrelated sectors.
Separate facts and assumptions	Tag observed data, derived calculations, and scenario assumptions separately. Keep the deterministic packet unchanged.	Treating a forecast as a fact.
Study cash flow and balance-sheet risk	Reconcile profit with operating cash flow; review leverage, refinancing, dilution, related-party exposure, and contingent liabilities.	Buying earnings growth that is not supported by cash or funding quality.
Model downside before upside	Set Bear, Base, and Bull drivers, probabilities, invalidation points, and a margin-of-safety requirement before considering an entry.	Focusing on a target price without loss capacity.
Respect liquidity and sizing	Check turnover and tradability; size exposure from risk capacity and portfolio concentration, not only conviction.	A position that cannot be exited under stress.
Rerun on real catalysts	Update after results, guidance, material disclosures, corporate actions, or a thesis-breaking event. Compare the new packet with the prior run.	Reacting to noise while ignoring changed fundamentals.

Pre-run checklist

- Is the ticker, reporting period, and data as-of timestamp correct?
- What is the thesis, horizon, and explicit invalidation condition?
- Which official filing supports each material claim?
- Are peer companies and valuation methods appropriate for the sector?
- What happens to value under lower growth, weaker margins, higher rates, or currency pressure?
- What evidence is missing, stale, inconsistent, or provider-limited?
- What position size keeps the Bear case within the user's risk capacity?
- What future event should trigger the next task run?

Indonesia-specific evidence starting points

Use the IDX listed-company financial and annual report portal

(<https://www.idx.co.id/id/perusahaan-tercatat/laporan-keuangan-dan-tahunan>), issuer investor-relations pages, and OJK public information (<https://www.ojk.go.id>). Confirm publication dates and retain source links in the research record.

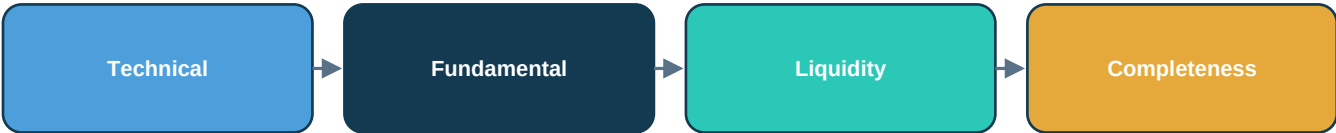
8 Scenario, Confidence, and Risk Contracts

Output	Contract
Bear/Base/Bull	Every scenario stores target price, return, and probability.
Expected return	Probability-weighted scenario return; it is not a guaranteed forecast.
Bear downside	Return from current price to bear target; used as an explicit risk gate.
Confidence	Composite of technical, fundamental, liquidity and completeness dimensions.
Hard block	A policy state that prevents a stronger action regardless of narrative.
Capital context	Input carried into the research packet; it does not execute allocation or orders.

Required interpretation text

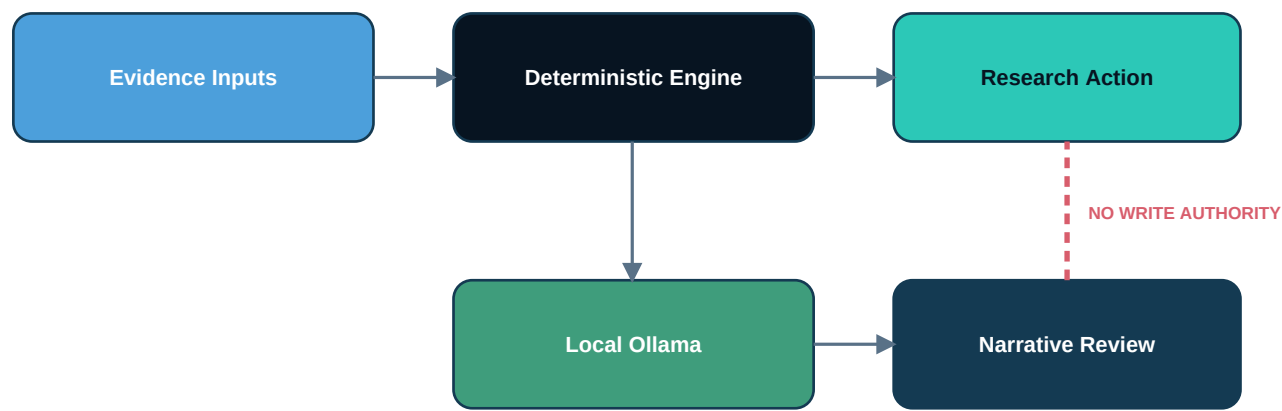
Every scenario view must label values as scenarios, display the as-of timestamp, preserve the current price used, and show evidence-completeness limitations. The UI must never display a target price without its downside context and policy status.

Figure 6. Confidence composition



9 AI Review Requirements

Figure 7. AI authority boundary



AI may	AI may not
Summarize the deterministic packet	Change ticker, price, timestamps, or source facts
List drivers and key risks	Recalculate scores or scenarios
Find contradictions and missing evidence	Override hard blocks or research action
Draft analyst follow-up questions	Approve publication or execute trades
Return unavailable status on failure	Cause deterministic work to be discarded

Model configuration

AI_PROVIDER=ollama; OLLAMA_BASE_URL=http://ollama:11434; OLLAMA_MODEL=qwen3:8b; no external AI key is required.

10 Evidence and Data Source Policy

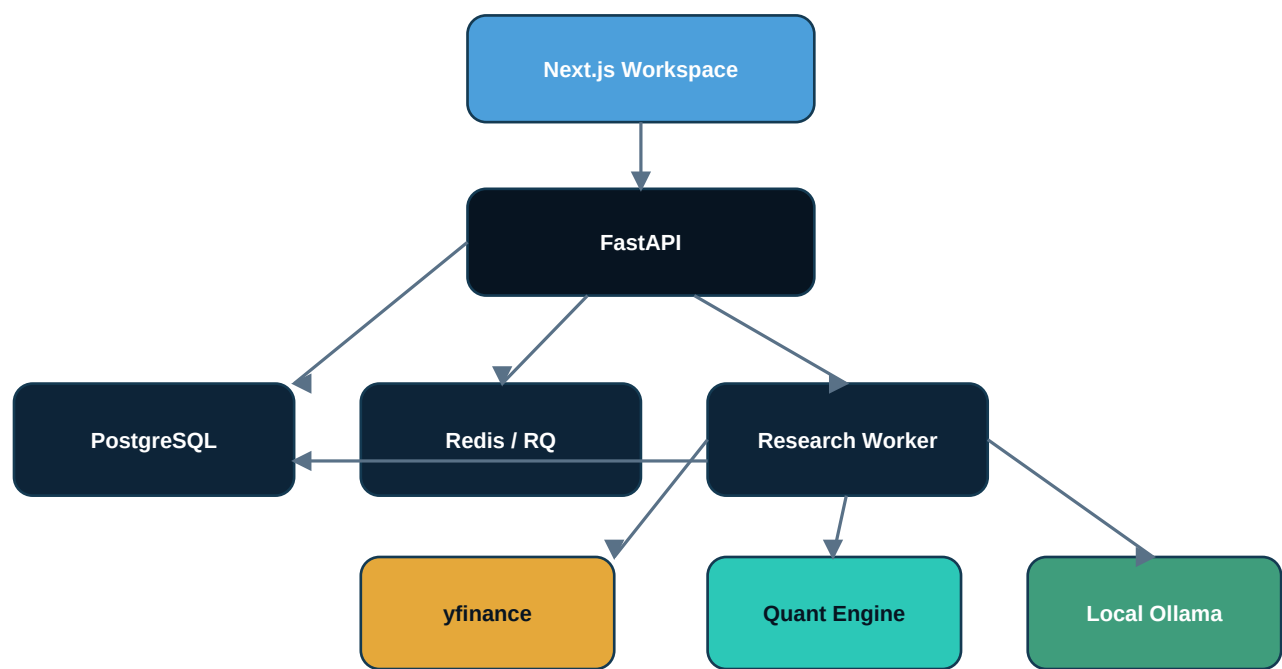
Source class	MVP source	Rights/quality posture	Failure behavior
Price and volume	Yahoo Finance via yfinance	Public/unofficial; terms and continuity must be reviewed	Ticker result fails; no paid fallback
Fundamentals	Fields exposed through yfinance	May be incomplete and not filing-normalized	Missing fields reduce coverage
AI	Local Ollama	Data remains local to the configured host	Deterministic result still publishes
Future filings	IDX/OJK/issuer IR	Public availability does not automatically grant bulk automation rights	Not implemented in v0.2.1
Future licensed feed	IDX/vendor adapter	Entitlement required	Architecture extension; not automatic fallback

Evidence record minimum

- evidence_type
- source_name and source_uri
- observed/retrieved timestamp
- ticker in payload
- rights note
- content hash
- run ID linkage

11 Technical Architecture

Figure 8. Runtime architecture



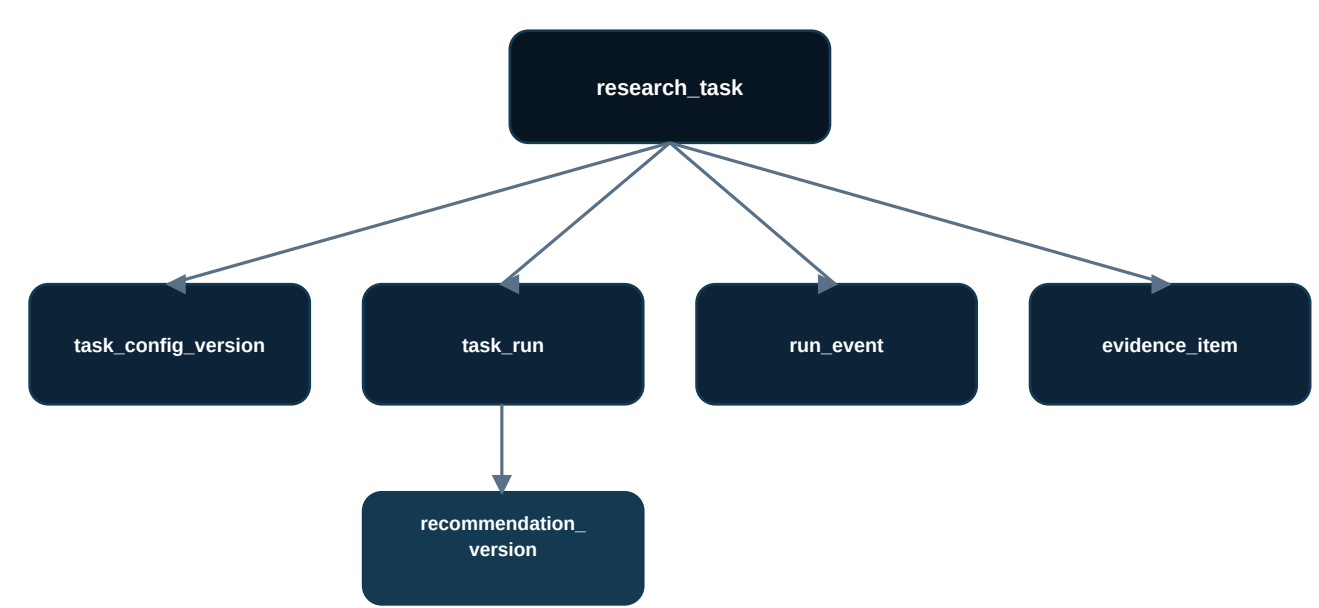
Component	Technology	Responsibility
Web	Next.js 15.5.24 + React 19.1.5	Task workspace, polling, drill-down
API	Python 3.12 + FastAPI/Pydantic	Validation, task/config APIs, queue submission, read models
Worker	Python + RQ	Multi-ticker orchestration, events, publication
Database	PostgreSQL	Tasks, versions, runs, events, packets, evidence
Queue	Redis	Asynchronous job delivery
Market adapter	yfinance	Free-only .JK EOD data
AI adapter	Ollama /api/chat	Local narrative review

Architecture style

Contract-enforced modular monolith. API and worker share domain code but run as independent containers. This preserves low operating cost and simple local deployment while keeping provider, policy, and orchestration seams replaceable.

12 Data Model and Immutability

Figure 10. Persistence model



Entity	Key fields	Mutability
research_task	id, ticker, horizon_days, capital_idr, params_json	Stable identity; active config projection may change
task_config_version	task_id, version, config_hash, config_json	Append-only
task_run	task_id, status, result_json, timestamps	State transitions then immutable terminal packet
run_event	run_id, stage, progress, message, created_at	Append-only
recommendation_version	run_id, version, action, content_hash, payload	Append-only
evidence_item	run_id, type, source, payload, hash	Append-only

Backward compatibility

The existing `research_task` and `task_run` schemas are not destructively altered. Rich task fields remain in `params_json`. New tables are created by metadata initialization. Legacy single-ticker tasks receive a fallback version-1 projection until they are updated.

13 API Contract

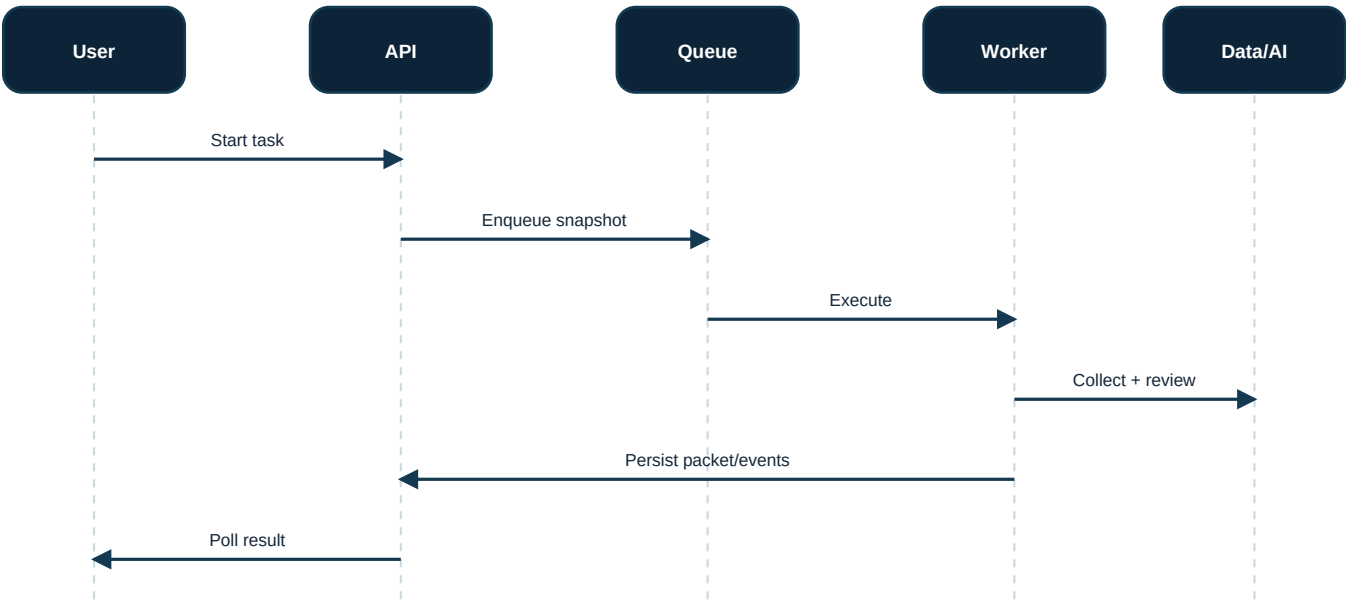
Method	Path	Purpose	Key response
GET	/v1/dashboard	Workspace summary	Counts, recent tasks and runs
GET	/v1/research-tasks	List active tasks	Task cards with latest run
POST	/v1/research-tasks	Create task/config v1	201 task response
GET	/v1/research-tasks/{id}	Get task	Config identity and latest run
PATCH	/v1/research-tasks/{id}	Publish new config	Incremented version/hash
DELETE	/v1/research-tasks/{id}	Archive task	Archived state or 409
POST	/v1/research-tasks/{id}/runs	Start run	202 queued run or 409
GET	/v1/research-tasks/{id}/runs	Task history	Ordered runs
GET	/v1/runs	Global/filterable history	Ordered runs
GET	/v1/runs/{id}	Poll run	Stage, progress, events, packet

Compatibility rule

POST /v1/research-tasks continues to accept the v0.1 ticker field. The service converts it to a one-element tickers array and generates a default task name.

14 Execution Sequence and Failure Semantics

Figure 11. Run execution sequence



Failure	Visible outcome	Recovery
Redis unavailable during start	503; run becomes FAILED with event	Restore queue and start a new run
One ticker provider failure	symbol_failed event; remaining tickers continue	PARTIAL packet; rerun later
All tickers fail	FAILED packet with per-ticker errors	Review provider status and rerun
Ollama unavailable	AI review marked unavailable	Deterministic packet remains publishable
Duplicate start	409 with active run ID	Open existing run instead
Worker process crash	Run may remain RUNNING in current MVP	Operational restart; watchdog is next hardening step

15 Frontend Experience Requirements

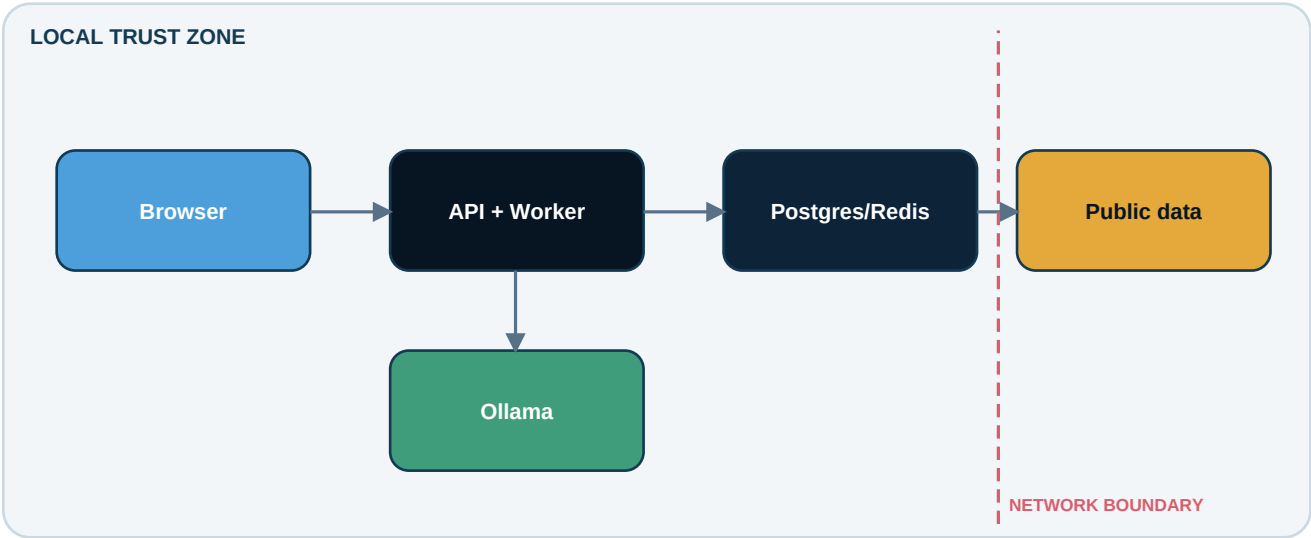
View	Must show	Must avoid
Overview	Task/run KPIs, recent tasks, recent runs, task CTA	Starting with a raw ticker form
Task Card	Config version, symbols, thesis, latest status/progress	Hiding whether the task is currently active
Task Detail	Thesis, universe, horizon, modules, config hash, run history	Editing terminal run outputs
Run Detail	Status, stage, progress, events, summary, ticker tabs	Showing a blank screen during background work
Analysis	Action, current price, upside/downside, confidence, scenarios	Presenting scenario target as guaranteed forecast
Evidence	Source link, evidence type, hashes	Unattributed numeric claims
Methodology	Deterministic/AI boundary and provider policy	Suggesting AI selects the action
Settings	Instant English/Bahasa Indonesia switch and active state	Reloading the page or mutating stored evidence

Responsive behavior

Desktop uses persistent sidebar and multi-column cards. Tablet collapses the sidebar label density. Mobile converts navigation to a horizontal strip and all analytical grids to one column. Tables may scroll horizontally but action buttons and status remain visible.

16 Security, Privacy, and Trust

Figure 13. Security trust zones



Control	v0.2.1	Production requirement
Authentication	None; local single user	OIDC/session identity and user-scoped tasks
Authorization	Process-local trust	Task/run ownership and role policies
Transport	Local HTTP defaults	TLS termination and secure headers
Secrets	.env and Docker environment	Secret store, rotation, no secrets in source
Input validation	Pydantic bounds and enums	Rate limits, body limits, abuse controls
AI privacy	Local Ollama	Document retention and prompt audit policy
Dependency security	Pinned lockfile and patched maintenance versions	Automated scanning and monthly updates

Threat priorities

- Unauthorized public access to a local research instance
- Malicious or malformed ticker/task payload
- Dependency vulnerabilities in the web runtime
- Provider content treated as executable or trusted narrative
- Cross-user data exposure after multi-user expansion
- Silent corruption of task config or evidence lineage

17 Non-Functional Requirements

Area	MVP target	Measurement
API availability	99% during local process uptime	Health endpoint and container state
Task start	202 or explicit error within 2 seconds excluding local host overload	API latency
Progress visibility	First worker event within 10 seconds after dequeue	Run event timestamps
Run duration	P95 under 5 minutes for one ticker with local model loaded	started_at to finished_at
Universe	Maximum 10 tickers	Pydantic validation
Determinism	Same facts/config/formula version yields same deterministic hash	Regression test
Recovery	PostgreSQL backup and documented restore	Restore rehearsal
Accessibility	Keyboard-operable forms and readable contrast	Manual audit; WCAG test in P1
Localization	English and Bahasa Indonesia UI; browser persistence	Translation QA and locale-aware content policy

18 Testing and Verification Strategy

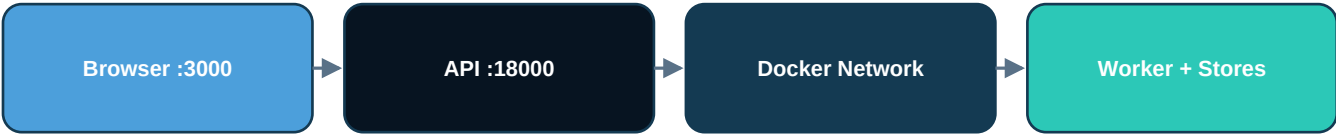
Layer	Coverage
Unit	Ticker normalization, policy gates, scoring, scenarios, schema validation, progress callbacks
Domain/API	Multi-ticker create, config version increment, hash persistence, legacy payload
Worker	Complete, partial, failed, event ordering, config snapshot, AI failure
Frontend	Production TypeScript build, task creation, polling, terminal packet rendering, EN/ID switching and persistence
Contract	OpenAPI response fields and backward-compatible create request
Integration	Postgres + Redis + worker + Ollama Compose smoke task
Operational	Backup/restore, container restart, provider outage, port conflict
Security	Dependency audit, secret scan, input abuse, unauthenticated exposure warning

Definition of Done for v0.2.1

- All backend tests and static checks pass.
- Frontend production build passes with lockfile.
- Shell scripts pass syntax validation.
- PDF renders without clipping or broken diagrams.
- Repository ZIP passes integrity test.
- README and implementation status do not claim automatic scheduling or report generation.
- GitHub is not modified without explicit user authorization.

19 Deployment and Operations

Figure 12. Local deployment topology



Bootstrap

```
cp .env.example .env ./scripts/bootstrap.sh
```

Operational routines

Routine	Command	Expected result
Start	./scripts/start.sh	All services healthy
Smoke	./scripts/smoke-test.sh	Health and free quote checks
Demo task	./scripts/task-demo.sh	Published multi-stock packet
Tests	./scripts/test.sh	Backend tests and frontend build
Backup	./scripts/backup.sh	Timestamped database dump
Restore	./scripts/restore.sh	Database restored from selected backup
Stop	./scripts/stop.sh	Compose services stopped

20 Implementation Mapping

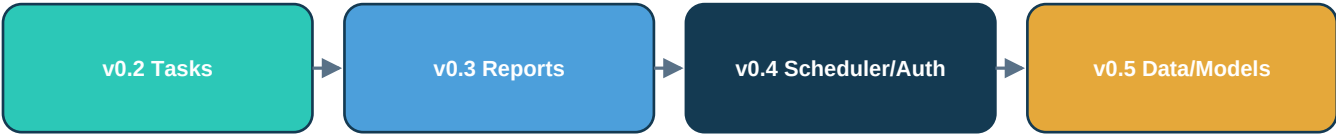
Requirement area	Repository location
Task/config APIs	backend/app/api/research.py
Task config and event persistence	backend/app/models.py
Request/response validation	backend/app/schemas.py
Run orchestration	backend/app/worker_jobs.py
Deterministic analysis progress	backend/app/services/analysis_engine.py
Task workspace UI	frontend/app/page.tsx
Interface localization	frontend/lib/i18n.ts
Visual system	frontend/app/globals.css
Browser API client	frontend/lib/api.ts
End-to-end demo	scripts/task-demo.sh
Architecture and contracts	docs/ARCHITECTURE.md and docs/TASK_ORCHESTRATION.md

Current implementation status

Implemented: task dashboard, multi-ticker tasks, configuration versions and hashes, snapshots, queue, events, progress, complete/partial/failed publication, per-ticker result views, persistent English/Bahasa Indonesia interface switching, local Ollama, free-only data, Docker and verification scripts. Not implemented: scheduler, authentication, official filing normalization, sector valuation, automatic report generation, outcome ledger.

21 Roadmap and Delivery Epics

Figure 14. Delivery roadmap



Epic	Outcome	Exit gate
E1 Task orchestration	Reusable task, config version, run trace, EN/ID interface	Implemented in v0.2.1
E2 Run PDF report	Executive and audit-ready report per terminal run	Report hash and download API
E3 Scheduler	Due-task execution with idempotency and missed-run recovery	No duplicate scheduled packets
E4 Identity	User-scoped tasks and role-aware access	Cross-user isolation tests
E5 Official evidence	IDX/OJK/issuer filing ingestion	Point-in-time lineage and rights review
E6 Sector routing	Bank, insurer, commodity, consumer and infrastructure models	Model-fit rules and tests
E7 Outcome ledger	Compare recommendations with realized outcomes	Point-in-time, survivorship-aware evaluation
E8 Licensed provider	Replaceable entitled market-data adapter	Contract parity and entitlement enforcement

22 Acceptance Scenarios

Scenario	Given / When	Then
Create task	Three valid IDX tickers	Symbols normalize, config v1 and hash return
Legacy create	ticker=BBCA only	One-symbol task is created
Duplicate ticker	BBCA, BBRI, BBKA	Universe becomes BBKA.JK and BBRI.JK
Run task	No active run	202, queued event, immutable snapshot
Duplicate run	Run already queued/running	409 and existing run ID
Partial data	One of three tickers fails	Two analyses publish; status PARTIAL
Ollama offline	Deterministic work succeeds	Packet publishes with AI unavailable
Config update	Horizon changes	Version increments, new hash; old run snapshot unchanged
Archive active task	Run still active	409; no archive
Free-only guard	Paid provider configured	Startup rejects configuration

23 Risks and Mitigations

Risk	Impact	Mitigation
Unofficial provider availability	Missing or rate-limited runs	Explicit failure, caching, future licensed adapter
Fundamental field inconsistency	Misleading cross-company comparison	Completeness score, field provenance, future filing normalization
Generic valuation logic	Poor model fit by sector	Label current scenarios; implement sector routing before broad use
Long local AI latency	Slow multi-ticker run	Progress events, smaller model option, AI non-blocking future mode
Stuck worker state	Run appears permanently active	Add heartbeat/watchdog and retry policy in next hardening
No authentication	Unsafe public exposure	Local-only warning; add identity before shared deployment
False certainty	User over-trusts action	Research labels, evidence gaps, scenario language, no trade execution

24 Glossary

Term	Definition
Task	Reusable Indonesian equity research mandate
Config version	Immutable numbered task-method configuration
Run	One asynchronous execution of a task snapshot
Run event	Append-only stage/progress observation
Evidence packet	Published task-level bundle containing analyses, source lineage and hashes
Research action	Deterministic decision-support category; not an order
Partial	At least one ticker succeeded and at least one failed
Free-only	No automatic billable market-data provider path exists
Local Ollama	Model inference served from the user's environment
PIT	Point in time: only information available by the evaluation cutoff

25 References and Approval Checklist

Repository and implementation references

- BuktiSaham repository: <https://github.com/radhian/buktisaham>
- IDX listed-company financial and annual reports: <https://www.idx.co.id/id/perusahaan-tercatat/laporan-keuangan-dan-tahunan>
- OJK public information and investor education: <https://www.ojk.go.id>
- Next.js security maintenance guidance: <https://nextjs.org/blog/tag/security>
- Ollama local model API: <https://ollama.com>
- yfinance project and data-use notes: <https://github.com/ranaroussi/yfinance>
- IDX data services for future licensed production integration: <https://data.idx.co.id>

Approval checklist

- Product approves task-first workflow and MVP non-goals.
- Research owner approves deterministic policy terminology and scenario labeling.
- Engineering approves API, event, data model and compatibility contracts.
- Security accepts local-only boundary for MVP and blocks public exposure without identity controls.
- Data owner accepts free-only provider limitations for MVP.
- Delivery owner accepts the roadmap order: report, scheduler, identity, official evidence, sector routing.

Approval statement

Approval of this document authorizes implementation and iteration against the stated v2.1 contracts. It does not authorize brokerage execution, public multi-user deployment, paid data procurement, or claims of investment performance.